

Linchen Qian

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INTRODUCTION

I am currently a Ph.D. candidate in Computer Science at the University of Miami, focusing on deep learning for medical image analysis, with an emphasis on attention-based and VAE-based deep neural networks for geometry reconstruction from MRI and CT.

EDUCATION

- **University of Miami** 2020 – Present
Ph.D. in Computer Science Coral Gables, FL, USA
 - Research Interest: deep learning for medical image analysis (e.g., lumbar spine, aortic valve)
- **University of Southern California** 2017 – 2019
M.S. in Computer Science Los Angeles, CA, USA
- **Beihang University** 2013 – 2017
B.S. in Computer Science Beijing, China

PROGRAMMING LANGUAGES

- Python (including PyTorch, PyTorch Geometric, NumPy, pandas, scikit-learn, etc.), Java, C/C++, C#, SQL, etc.

RESEARCH EXPERIENCE

- **University of Miami, Department of Computer Science** 2020 – Present
Research Assistant Coral Gables, FL
 - **Uncertainty Modeling:** Designed CQ-VAE, a generative network with a coordinate-quantized discrete latent space that learns shape distributions from multi-annotator lumbar spine MR images, achieving a strong correlation between estimated uncertainty and true shape error.
 - **Artifact-Free Geometry Reconstruction:** Designed UNet-DeformSA and TransDeformer, attention-based shape-deformation networks that reconstruct lumbar spine meshes from the MR images with consistent inter-patient mesh correspondence.
 - **Quality Control:** Built a TransDeformer variant network for shape error estimation that ranks reconstructed geometries by predicted error, enabling prioritized clinician review.
 - **Aortic Valve Modeling:** Introduced a template-fitting remeshing pipeline to produce high-quality structured quad meshes on aortic valve with simplified loss functions.
- **USC Information Sciences Institute** 2018
Research Assistant Marina del Rey, CA
 - Extracted the optimal hypothesis from a Knowledge Graph where ontological constraints were integrated to resolve inconsistencies in entity-relationship statements

SELECTED PROJECTS

- **CQ-VAE: Coordinate-Quantized VAE for Shape Uncertainty Estimation**
 - Formulated shape uncertainty estimation on lumbar spine MR images, motivated by inter-annotator disagreement on lumbar spine disc boundaries
 - Designed a coordinate-quantized variational autoencoder, **CQ-VAE**, with a discrete latent space, trained with a global-greedy shape sampling-matching strategy that balances generative diversity against accuracy on existing ground truth shapes.
 - Demonstrated a strong correlation between estimated shape variation and ground truth bias.
- **Attention-Based Shape-Deformation Networks for Lumbar Spine MR Imaging**
 - Reconstructed artifact-free lumbar spine meshes with inter-patient correspondence, enabling consistent and accurate measurement of medical parameters related to disc degeneration.
 - Designed two novel attention-based shape-deformation networks, **UNet-DeformSA** and **TransDeformer**, which integrate self- and cross-modal attention to capture long-range dependencies between mesh and image features.
 - Proposed a three-mode training strategy that ensures robust deformation under imperfect template initialization.
 - Built a **TransDeformer** variant for point-wise shape error estimation, enabling reconstructed geometries to be ranked by predicted error for prioritized clinician review.
 - Demonstrated state-of-the-art accuracy in lumbar spine geometry reconstruction and downstream medical parameter measurement
- **Structured Quad-Mesh Reconstruction of the Aortic Valve from 3D CT**

- Developed a template-fitting remeshing algorithm during data preprocessing to produce annotated datasets of high-quality structured quad meshes for the aortic valve, with consistent inter-patient mesh correspondence.
- Simplified the training process to only two loss terms while preserving mesh smoothness and element quality, replacing the complicated loss functions required by unstructured triangular meshes.
- Demonstrated improved smoothness, element quality, and reconstruction accuracy of aortic valve surface meshes, achieving a 20–25% improvement in some metrics over unstructured triangular meshes.

SELECTED REVIEWER EXPERIENCE

- Expert Systems with Applications
- IEEE Journal of Biomedical and Health Informatics (JBHI)
- Neuroscience Informatics

PUBLICATIONS

BOLD = FIRST AUTHOR

First-Author Publications

- [1] **L. Qian**, J. Chen, L. Ma, T. Urakov, W. Gu, and L. Liang. **Attention-based Shape-Deformation Networks for Artifact-Free Geometry Reconstruction of Lumbar Spine from MR Images**. *IEEE Transactions on Medical Imaging*, 2025. [\[Code\]](#)
- [2] **L. Qian**, J. Chen, R. Gong, W. Sun, M. Liu, and L. Liang. **Shape Deformation Networks for Automated Aortic Valve Finite Element Meshing from 3D CT Images**. *SPIE Medical Imaging*, 2026.
- [3] **L. Qian**, J. Chen, L. Ma, T. Urakov, W. Gu, and L. Liang. **A Novel Attention-Based Network for Geometry Reconstruction with Error Estimation from Medical Images**. *SPIE Medical Imaging*, 2025. [\[Code\]](#)
- [4] **L. Qian**, J. Chen, L. Ma, T. Urakov, and L. Liang. **A Sequential Geometry-Reconstruction-Based Deep Learning Approach to Improve Accuracy and Consistence of Lumbar Spine MRI Image Segmentation**. *SPIE Medical Imaging*, 2024. [\[Code\]](#)
- [5] **L. Qian**, J. Chen, T. Urakov, W. Gu, and L. Liang. **CQ-VAE: Coordinate Quantized VAE for Uncertainty Estimation with Application to Disk Shape Analysis from Lumbar Spine MRI Images**. *Proc. Int. Conf. Machine Learning and Applications (ICMLA)*, 2020. [\[Code\]](#)

Co-Author Publications

- [6] J. Chen, **L. Qian**, L. Ma, T. Urakov, W. Gu, and L. Liang. **SymTC: A Symbiotic Transformer-CNN Net for Instance Segmentation of Lumbar Spine MRI**. *Computers in Biology and Medicine*, 2024.
- [7] J. Chen, **L. Qian**, R. Gong, C. Sun, T. Qin, T. Pham, C. Martin, M. Zafar, J. Elefteriades, W. Sun, and L. Liang. **A Computational Pipeline for Patient-Specific Modeling of Thoracic Aortic Aneurysm: From Medical Image to Finite Element Analysis**. *SPIE Medical Imaging*, 2026.
- [8] L. Ma, J. Chen, **L. Qian**, and L. Liang. **A General Approach to Improve Adversarial Robustness of DNNs for Medical Image Segmentation and Detection**. *SPIE Medical Imaging*, 2024.
- [9] J. Chen, **L. Qian**, T. Urakov, W. Gu, and L. Liang. **Adversarial Robustness Study of Convolutional Neural Network for Lumbar Disk Shape Reconstruction from MR Images**. *SPIE Medical Imaging*, 2021.